

The sharp constant in the shift-wall bridge for odd-cycle polynomials

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Abstract

Let

$$M_0 = 1, \quad M_1 = 1 + z, \quad M_n = M_{n-1} + n^2 z M_{n-2},$$

and write $L(f)_q = f_q^2 - f_{q-1}f_{q+1}$ for the coefficientwise Turán operator. A previous square-tail theorem implies

$$(2r)^2 L((1+z)M_{2r-2})_q \geq 2L(M_{2r-1})_q, \quad 0 \leq q \leq r-2.$$

We determine the best possible replacement for the factor 2. Put

$$\Omega_t(r) = \frac{(2r)^2 L((1+z)M_{2r-2})_{r-t}}{2L(M_{2r-1})_{r-t}}, \quad r \geq t \geq 2.$$

Then

$$\Omega_t(r) \geq \Omega_2(1350) = 1.327509156956680599863203087838 \dots,$$

with equality only at $(r, t) = (1350, 2)$. Consequently the sharp constant in the unnormalized bridge is

$$2\Omega_2(1350) = 2.655018313913361199726406175676 \dots$$

The proof has one exceptional stripe and one uniform barrier. The $t = 2$ minimum is certified by an exact prefix and a positive-coefficient half-line certificate. A lossless telescoping transfer sends every fixed-column lower bound for the square-tail quotient ρ to the corresponding bridge stripe. The stripe $t = 3$ is treated directly, while $\rho > 4/3$ closes every stripe $t \geq 4$. For $t \geq 5$ this barrier is assembled from terminal identities, a 123,753-cell exact core, six fixed-column tail certificates, a re-anchored top estimate, and an all-parameter joint certificate; on that region it is also a uniform improvement, by a factor $1.314 \dots$, on the sharp square-tail constant of the earlier paper. A byproduct localizes the whole question: every cell below $4/3$ lies on the stripe $t = 2$ with $r \leq 3000$. Every decision gate is an integer or rational comparison; decimal values are reported only for orientation. The stripes $t = 9$ and $t = 10$ each carry a second, separately gated proof chain, one of which rests on an exact two-leg certificate for a quantitative decrement shell.

1 The extremal question

For a polynomial $f(z) = \sum_k f_k z^k$, put

$$L(f)_q = f_q^2 - f_{q-1}f_{q+1}, \tag{1}$$

where coefficients outside the row are zero. The wall recurrence

$$M_0 = 1, \quad M_1 = 1 + z, \quad M_n = M_{n-1} + n^2 z M_{n-2} \tag{2}$$

is the even-variable form of the odd-cycle continuant. The rows are real-rooted with positive coefficients, so the Turán forms used below are positive on their natural range; the determinant, spectral, and permutation interpretations are developed in [3].

For $r \geq 2$ and $0 \leq q \leq r - 2$, define the raw bridge ratio

$$\mathcal{B}(r, q) = \frac{(2r)^2 L((1+z)M_{2r-2})_q}{L(M_{2r-1})_q}.$$

Writing $t = r - q$ reverses the coefficient index and gives

$$\Omega_t(r) = \frac{\mathcal{B}(r, r-t)}{2} = \frac{(2r)^2 L((1+z)M_{2r-2})_{r-t}}{2L(M_{2r-1})_{r-t}}, \quad r \geq t \geq 2. \quad (3)$$

The bridge theorem in [3] is $\Omega_t(r) \geq 1$. Its constant 2 is structural: it is the amount paid by the square-tail inequality and the Turán triangle. The present question is extremal: what is $\inf_{r \geq t \geq 2} \Omega_t(r)$?

Set

$$\omega_{\text{wall}} = \Omega_2(1350), \quad K_{\text{wall}} = 2\omega_{\text{wall}}. \quad (4)$$

Theorem 1.1 (sharp shift-wall constant). *For all integers $r \geq t \geq 2$,*

$$\Omega_t(r) \geq \omega_{\text{wall}} = 1.327509156956680599863203087838 \dots, \quad (5)$$

with equality if and only if $(r, t) = (1350, 2)$. Equivalently,

$$(2r)^2 L((1+z)M_{2r-2})_q \geq K_{\text{wall}} L(M_{2r-1})_q, \quad 0 \leq q \leq r - 2, \quad (6)$$

where

$$K_{\text{wall}} = 2.655018313913361199726406175676 \dots$$

The constant is best possible, and equality in (6) occurs only at $(r, q) = (1350, 1348)$.

The exact value in (5) is a reduced rational. Its numerator has 6,733 decimal digits, its denominator 6,732, and the SHA-256 digest of the canonical ASCII string `numerator/denominator` is

$$2\text{fc6689fd42dd6eaa29088aeb0077e1120878fd2f6b54cf2e8dbd50e823a5691}.$$

The decimal is not used as a proof target.

The proof is organized around the rational separator

$$\omega_{\text{wall}} < \frac{4}{3}, \quad \frac{4}{3} - \omega_{\text{wall}} = 0.005824176376652733470130 \dots \quad (7)$$

The exceptional stripe $t = 2$ supplies the sharp value. Every stripe $t \geq 3$ is strictly above $4/3$. The main work is therefore not a decimal comparison with (5), but the uniform exact theorem $\rho > 4/3$ proved in Section 6.

1.1 Place in the four-paper program

The logical input is the square-tail theorem of [3]. That paper builds the odd-cycle, Jacobi, and coefficient-curvature machinery and proves the bridge with the structural factor 2; this paper determines the exact constant left open there. It also left a prediction: its own sharp square-tail constant $C_\star = 2\rho(129, 2) = 2.0291523 \dots$ propagates through the bridge, but is not sharp there, and the bridge's extremal cell is *not* the image of $(129, 2)$. Theorem 1.1 confirms both. The extremal cell is $(1350, 2)$, and K_{wall} exceeds $C_\star$ by 30.8% and the structural factor 2 by 32.8%. The sharp multiplicative Delannoy theorem in [1] is the external application that first produced the bridge question. The grounded-path algorithms of [2] expose a related determinantal mechanism, but are a conceptual sibling rather than a premise of Theorem 1.1.

2 The square-tail quotient and the wall level

Reverse and normalize the even wall row by

$$\Pi_r(u) = \frac{u^{r-1} M_{2r-2}(1/u)}{(2r-1)!}. \quad (8)$$

For $r \geq 2$ and $2 \leq t \leq r$, define

$$S_{r,t} = L((1+u)\Pi_r)_t, \quad V_{r,t} = L(\Pi_r)_{t-1}, \quad (9)$$

$$d_{r,t} = S_{r,t} - \left(\frac{2r-1}{2r}\right)^2 S_{r-1,t}, \quad \rho(r,t) = \frac{r^2 d_{r,t}^2}{2S_{r,t} V_{r,t}}. \quad (10)$$

The results of [3] give $S_{r,t}, V_{r,t}, d_{r,t} > 0$ and $\rho(r,t) \geq 1$ on this domain. The quotient ρ measures the exact surplus in the cross-row square-tail inequality. It is not the bridge ratio Ω : the next section records the one-way transfer between them.

The unbounded part of the ρ -array is organized by one increasing scalar. Let

$$\hat{\Pi}_r(u) = \frac{\Pi_r(u)}{\Pi_r(0)}, \quad \mathbf{a}_r = 6[u]\hat{\Pi}_r(u).$$

The cancellation-free Wallis ladder in [3] proves that $\mathbf{a}_r$ is strictly increasing. Its role is geometric. For fixed t , the inequality

$$\mathbf{a}_r \leq (t-2)^2 \quad (11)$$

defines the top region; its complement is the joint region. The proof never uses a floating approximation to decide which side of (11) contains a cell. The few crossings needed below are certified by outward rational intervals.

3 A lossless fixed-column transfer

The bridge in [3] was obtained by inserting the square-tail constant 1 at every row. Keeping the local quotient $\rho(j,t)$ instead produces a sharper inequality with no additional loss.

For $0 \leq t \leq r$, put

$$O_{r,t} = [z^{r-t}]M_{2r-1}, \quad E_{r,t} = [z^{r-t}]M_{2r},$$

and normalize

$$P_r(u) = \sum_t \frac{O_{r,t}}{O_{r,0}} u^t, \quad Q_r(u) = \sum_t \frac{E_{r,t}}{E_{r,0}} u^t, \quad x_r = \frac{E_{r-1,0}}{O_{r,0}}.$$

The wall recurrence gives

$$P_r = P_{r-1} + u x_r Q_{r-1}. \quad (12)$$

For a fixed $t \geq 2$, set

$$\begin{aligned} p_j &= \sqrt{L(P_j)_t}, \\ A_j &= j x_j \sqrt{L((1+u)Q_{j-1})_t}, \\ v_j &= x_j \sqrt{L(Q_{j-1})_{t-1}}. \end{aligned}$$

All three depend on t , which is suppressed from the notation.

Lemma 3.1 (lossless transfer). *For every pair of integers $r \geq t \geq 2$, there are positive weights*

$$\lambda_{j,t} = \frac{A_j - A_{j-1}}{A_r}, \quad t \leq j \leq r, \quad \sum_{j=t}^r \lambda_{j,t} = 1,$$

such that

$$\Omega_t(r) \geq \left(\sum_{j=t}^r \frac{\lambda_{j,t}}{\sqrt{\rho(j,t)}} \right)^{-2} \geq \min_{t \leq j \leq r} \rho(j,t). \quad (13)$$

Proof. Two facts from [3] carry the argument. First, if f and g are real-rooted with non-negative coefficients and lie in *proper position*, then polarizing the Newton inequality for the pencil $f + vg$ gives the Turán triangle

$$\sqrt{L(f+g)_t} \leq \sqrt{L(f)_t} + \sqrt{L(g)_t}. \quad (14)$$

The position hypothesis is not removable, and the minimal witness is integral: $f = (1, 2, 1)$ and $g = (1, 3, 1)$ are both real-rooted with positive coefficients, yet at $t = 1$ they give $\sqrt{21} > \sqrt{3} + \sqrt{8}$, which on squaring twice is the integer statement $100 > 96$. What fails is position alone — the double root of f at -1 lies strictly between the two roots of g . Second, the two rows appearing in (12) do lie in proper position. The zeros of P_{j-1} are $-\lambda_i^2$ and those of uQ_{j-1} are the $-\nu_i^2$ together with 0, where λ_i and ν_i run over the positive spectra of the two consecutive Jacobi matrices of [3]; strict Cauchy interlacing and spectral symmetry give

$$0 < \lambda_1 < \nu_1 < \cdots < \lambda_{j-1} < \nu_{j-1},$$

so the zeros strictly interlace. Applying (14) to (12) therefore gives the Turán triangle

$$p_j \leq p_{j-1} + v_j. \quad (15)$$

In the same normalization, $A_j^2 - A_{j-1}^2$ is the oriented decrement and

$$\rho(j,t) = \frac{(A_j^2 - A_{j-1}^2)^2}{2A_j^2 v_j^2}.$$

The decrement is positive, so

$$A_j - A_{j-1} = \frac{A_j^2 - A_{j-1}^2}{A_j + A_{j-1}} \geq \sqrt{\frac{\rho(j,t)}{2}} v_j. \quad (16)$$

Both p_{t-1} and A_{t-1} vanish. Iterating (15), using (16), and then telescoping gives

$$p_r \leq \sum_{j=t}^r v_j \leq \sqrt{2} \sum_{j=t}^r \frac{A_j - A_{j-1}}{\sqrt{\rho(j,t)}} = \sqrt{2} A_r \sum_{j=t}^r \frac{\lambda_{j,t}}{\sqrt{\rho(j,t)}}.$$

Reversal in (3) is exactly $\Omega_t(r) = 2A_r^2/p_r^2$. This proves the first inequality in (13). The second follows because the $\lambda_{j,t}$ are positive and sum to one. $\square$

Thus a lower bound valid on an entire ρ -column transfers to the bridge at the same constant. The direction is important: no monotonicity of $\Omega_t(r)$ in t is asserted or needed.

4 The exceptional stripe

The sharp value occurs on $t = 2$, so this stripe is proved directly from the bridge ratio rather than through Lemma 3.1.

Put $m = r - 1$ and introduce the Wallis ladder of [3],

$$h_j = \left(\frac{(2j)!!}{(2j-1)!!} \right)^2, \quad T_j = 1 + \sum_{i=1}^j h_i^{-1}, \quad v_m = \frac{h_m}{(2m+1)^2},$$

whose rows are determined by $B_0(j) = T_j$, $C_0(j) = 1$,

$$B_k(j) = B_k(j-1) + h_j^{-1} C_k(j-1), \quad C_k(j) = C_k(j-1) + \frac{h_j}{(2j+1)^2} B_{k-1}(j), \quad (17)$$

with seeds $B_k(0) = 0$ for $k \geq 1$, $C_1(0) = 1$, and $C_k(0) = 0$ for $k \geq 2$.

The ladder is a rewriting of the two normalized rows of §3. On the even side this is proved in [3], in the equivalent form $B_k(j)/T_j = [u^k]\Pi_{j+1}/[u^0]\Pi_{j+1}$:

$$\frac{B_k(j)}{T_j} = [u^k]\mathbf{Q}_j. \quad (18)$$

The odd side is the companion statement. We record it because it is what makes (21) a rewriting rather than a numerical coincidence:

$$C_k(m) = [u^k]\mathbf{P}_r = \frac{[z^{r-k}]M_{2r-1}}{[z^r]M_{2r-1}}. \quad (19)$$

For (19), the wall recurrence first gives

$$x_{j+1} = \frac{h_j T_j}{(2j+1)^2}. \quad (20)$$

Now $C_0(j) = 1 = [u^0]\mathbf{P}_{j+1}$, and the coefficient of u^k in (12) reads

$$[u^k]\mathbf{P}_{j+1} = [u^k]\mathbf{P}_j + x_{j+1}[u^{k-1}]\mathbf{Q}_j.$$

By (18) and (20) the last term equals $h_j(2j+1)^{-2}B_{k-1}(j)$, so this recursion is exactly the second half of (17). Induction on j proves (19).

Substituting (18) and (19) into (3) and cancelling the common normalizations gives

$$\Omega_t(r) = 2r^2 v_m^2 \frac{P_t(B(m))}{Q_t(C(m))}, \quad (21)$$

where

$$\begin{aligned} P_t(B) &= (B_{t-1} + B_t)^2 - (B_{t-2} + B_{t-1})(B_t + B_{t+1}), \\ Q_t(C) &= C_t^2 - C_{t-1}C_{t+1}. \end{aligned}$$

Equation (21) was also verified directly against the raw wall recurrence in exact rational arithmetic at every cell with $2 \leq t \leq r \leq 9$, and against the rational ladder at the extremal row.

Proposition 4.1 (the $t = 2$ minimum). *For every integer $r \geq 2$,*

$$\Omega_2(r) \geq \Omega_2(1350),$$

with equality only at $r = 1350$.

Exact certificate. The prefix computes all 3,000 cells $2 \leq r \leq 3001$ as reduced rational comparisons with $\Omega_2(1350)$. Every numerator and denominator is positive, the equality set is $\{1350\}$, and all other cells are strictly larger. The nearest competitor is $r = 1349$, whose positive gap is

$$3.473032420463471 \dots \times 10^{-9}.$$

The comparison stream has SHA-256 digest

$$943bebf b449b9ec178562598433b1f76dd9f3cb0cab0e8203bfce784882245ae.$$

For the half-line, set $b_k = (\pi/2)B_k$ and $x = \tau_m - \tau_{3000}^{\text{lo}} \geq 0$. Outward rational envelopes give

$$P_2(b) \geq P_{\text{lo}}(x), \quad 0 < Q_{\text{lo}}(x) \leq Q_2(C) \leq Q_{\text{hi}}(x).$$

The exact Wallis bound supplies a strict scalar factor $k_m > 1/2$. The degree-ten polynomial

$$H(x) = P_{\text{lo}}(x) - 2\Omega_2(1350)Q_{\text{hi}}(x)$$

has all eleven rational coefficients strictly positive. Its indexed coefficient digest is

$$d8ba995d1f52b231ff5e198bd881ec9d1bd0def96c8b9c1c6efbf1f95130482a.$$

Hence $\Omega_2(r) > \Omega_2(1350)$ for every $r \geq 3001$. The prefix and half-line certificate overlap at $r = 3001$. $\square$

The target rational itself is regenerated from the wall recurrence. It is not embedded as a several-thousand-digit constant. A one-percent upward mutation of the target makes the constant coefficient of H negative; this checks that the tail gate can detect a false strengthening.

The half-line certificate is stronger than Proposition 4.1 needs, and the surplus localizes the whole question.

Corollary 4.2 (localization). *The set of cells with $\Omega_t(r) < 4/3$ is exactly*

$$\{ (r, 2) : 808 \leq r \leq 2339 \},$$

a contiguous window of 1532 cells on the single stripe $t = 2$.

Proof. The coefficient capacity of H — the largest target for which every coefficient inequality still holds — is

$$1.337695687131883021756995 \dots > \frac{4}{3},$$

so the same construction at target $4/3$ retains eleven positive coefficients and gives $\Omega_2(r) > 4/3$ for every $r \geq 3001$. On the prefix $2 \leq r \leq 3001$ the exact comparisons that prove Proposition 4.1 also decide every cell against $4/3$: exactly 1532 lie below it, they are contiguous, and both boundaries are strict,

$$\Omega_2(807) > \frac{4}{3} > \Omega_2(808), \quad \Omega_2(2339) < \frac{4}{3} < \Omega_2(2340).$$

Every stripe $t \geq 3$ is strictly above $4/3$: Proposition 5.1 for $t = 3, 4$, and Theorem 6.1 with Lemma 3.1 for $t \geq 5$. $\square$

The tail certificate is not tight at its own left edge, since $\Omega_2(3001) = 1.3394612629 \dots$; the capacity therefore lies below the value it certifies against, as a valid bound must, and above $4/3$, as Corollary 4.2 needs. The extremal cell $(1350, 2)$ lies inside the window, so the entire sub- $4/3$ part of the array is one finite segment of one stripe.

5 The first higher stripes

The separator $4/3$ first appears on $t = 3$.

Proposition 5.1. *For every integer $r \geq 3$, $\Omega_3(r) > 4/3$. For every integer $r \geq 4$, $\rho(r, 4) > 4/3$ and consequently $\Omega_4(r) > 4/3$.*

Exact certificates. For $t = 3$, the bridge ratio is checked directly on $3 \leq r \leq 3001$ and a degree-fourteen positive-coefficient polynomial proves the remaining half-line. Its coefficient capacity is $1.4458344930865687354\dots > 4/3$; changing the target to $3/2$ creates negative coefficients.

For $t = 4$, the definition-level prefix checks the 997 cells $4 \leq r \leq 1000$. At anchor 1000, a degree-thirty-eight polynomial has all 39 coefficients strictly positive at target $4/3$. Its coefficient capacity is $1.3417349730931779905\dots$, so the margin is thin but exact. Lemma 3.1 then gives the bridge statement. $\square$

The two certificate joins are summarized in Table 1.

Table 1: Exact certificates for the first higher stripes. Capacities are reporting decimals of exact rationals.

stripe	object	exact prefix	tail degree	capacity
$t = 3$	Ω_3	$3 \leq r \leq 3001$	14	$1.4458344930\dots$
$t = 4$	$\rho(\cdot, 4)$	$4 \leq r \leq 1000$	38	$1.3417349730\dots$

6 The uniform $4/3$ barrier

Theorem 6.1 (bulk barrier). *For every pair of integers $j \geq t \geq 5$,*

$$\rho(j, t) > \frac{4}{3}. \quad (22)$$

Remark 6.2. Theorem 6.1 is used below only as a separator, but it is a statement about the square-tail quotient in its own right. The sharp global constant of [3] is $\min \rho = \rho(129, 2) = 1.0145761645\dots$ over $2 \leq t \leq r$; on the region $t \geq 5$ the present theorem raises that to $\rho > 4/3$, a factor of $1.3141776\dots$. It does not claim that $4/3$ is approached on any stripe; see §8.

The proof partitions the array by mechanisms rather than by a large bounding box. Two terminal identities and one finite core handle $j < 504$. Columns 5 through 10 have exact positive-coefficient tails. Above them, the increasing wall level $\mathfrak{a}_j$ divides the array into top and joint regions.

6.1 Terminal cells and finite core

The upper two coefficient cells are symbolic identities from [3]:

$$\rho(j, j) = \frac{j^2}{2}, \quad \rho(j, j-1) > \frac{2j^2}{9}. \quad (23)$$

For a fixed column $t \geq 5$, these give

$$\rho(t, t) \geq \frac{25}{2}, \quad \rho(t+1, t) > 8.$$

The remaining cells below row 504 form the exact core

$$7 \leq j \leq 503, \quad 5 \leq t \leq j - 2. \quad (24)$$

There are $\sum_{j=7}^{503} (j - 6) = 123,753$ such cells. A factorial-cancelled integer formula tests

$$3N_{j,t}^2 - 4(2j^2 \widehat{S}_{j,t} \widehat{V}_{j,t}) > 0$$

at every cell, which is exactly $\rho(j, t) > 4/3$. There are no failed or degenerate cells. The minimum inside the box occurs at $(j, t) = (503, 5)$ and is

$$3.577109931649850292311076 \dots$$

The ordered comparison stream has digest

$$83797\text{ef}82\text{eca}71\text{c}6409\text{a}61\text{f}942\text{bd}81894\text{d}7352\text{cf}557\text{c}2\text{bae}2055\text{bd}40\text{b}9216402.$$

6.2 The fixed columns $5 \leq t \leq 8$

The unbounded certificate is uniform in form. At an anchor m_0 , exact outward envelopes in a variable $x \geq 0$ produce polynomials $u_{\text{lo}}, p_{\text{hi}}, q_{\text{hi}}, v$. The comparison polynomial is

$$H_{t,m_0}(x) = u_{\text{lo}}(x)^2 - \frac{4}{3} 8p_{\text{hi}}(x)q_{\text{hi}}(x)v(x)^2. \quad (25)$$

Strict positivity of every coefficient proves $\rho(j, t) > 4/3$ for all $j \geq m_0 + 1$. A definition-level integer prefix reaches the anchor and also checks the overlap cell $m_0 + 1$. Table 2 gives the exact joins. The capacity is the largest target allowed simultaneously by all coefficient inequalities, reported decimally only after the exact signs have been decided.

Table 2: Complete fixed-column certificates at target $4/3$.

t	exact prefix	tail starts	degree	capacity
5	$5 \leq j \leq 1000$	1001	46	1.410970990331...
6	$6 \leq j \leq 4006$	4007	54	1.518325000294...
7	$7 \leq j \leq 12398$	12399	62	1.581611335023...
8	$8 \leq j \leq 19999$	20000	70	1.420818931366...

Several smaller anchors fail, which is useful evidence about the mechanism: $m_0 = 2003$ fails for $t = 6$ with capacity $1.00047\dots$, and $m_0 = 6199$ fails for $t = 7$ with capacity $1.16994\dots$. These failures do not weaken the theorem; they locate where the chosen envelope becomes strong enough.

6.3 The top region

The top estimate already proved in [3] has more room than its original global anchor shows. On

$$j \geq 4, \quad 2 \leq t \leq j - 2, \quad \mathbf{a}_j \leq (t - 2)^2,$$

its proof gives

$$\rho(j, t) > \frac{1}{2} \left(1 - \frac{2}{4j - 3} \right)^2 \left(2 - \frac{1}{j} \right)^2. \quad (26)$$

Both factors increase with j . Restricting to $j \geq 504$ therefore yields the exact floor

$$\rho(j, t) > \frac{1}{2} \left(\frac{2011}{2013} \frac{1007}{504} \right)^2 = \frac{4100936855929}{2058631521408} = 1.9920693981\dots > \frac{4}{3}. \quad (27)$$

No new asymptotic estimate is needed: the earlier, much smaller floor was the same increasing product evaluated at $j = 4$.

6.4 The gate-compatible columns $t = 9, 10$

Columns 9 and 10 use the top region below an anchor and a positive-coefficient tail above it. Since $\mathbf{a}_j$ is increasing, the single inequality $\mathbf{a}_{m_0} \leq (t-2)^2$ places every row $504 \leq j \leq m_0$ in (27). Table 3 records the two joins.

Table 3: Prefix-free joins for columns 9 and 10. The finite core covers $j < 504$, the top theorem covers through m_0 , and the tail begins at $m_0 + 1$.

t	m_0	certified gate	tail starts	degree	capacity
9	68133	$\mathbf{a}_{68133} < 49$	68134	78	1.660681546638...
10	200000	$\mathbf{a}_{200000} < 64$	200001	86	1.515648188324...

For $t = 9$ the splice is exact at the level crossing: outward intervals prove

$$\mathbf{a}_{68133} < 49 < \mathbf{a}_{68134}.$$

Thus 68133 is the largest top-compatible anchor. A smaller build at 55000 fails, while the ceiling build passes with every one of its 79 coefficients positive. This corrects a tempting but false diagnostic: the envelope route is not dead for $t \geq 9$; it is inadequate only at anchors that sit too low. For $t = 10$, the build at 100000 fails and the build at 200000 passes. A separate sharpened-corridor Sturm certificate also closes the $t = 10$ joint range, but it is not used in the assembly. For $t = 9$, the sharpened-corridor slices combined with Proposition 8.1 give a second certificate chain across the dip $49 \leq \mathbf{a}_j \leq 768$. This chain is likewise corroborating rather than load-bearing.

6.5 The joint region for $t \geq 11$

It remains to prove the 4/3 floor when

$$j \geq 504, \quad t \geq 11, \quad \mathbf{a}_j > (t-2)^2. \quad (28)$$

The corridor reduction in [3] uses a real parameter Y and the rational function

$$\begin{aligned} G &= 2t(2t+1), & \alpha &= \frac{(t-1)(2t-1)}{t(2t+1)}, & \beta &= \frac{t(2t+1)}{(t+1)(2t+3)}, \\ x_0 &= \frac{Y-5}{G}, & u_0 &= \alpha \frac{Y-5}{Y}, & R_0 &= 1 + x_0 + (x_0 u_0 + x_0^2) \frac{1-\beta}{1-u_0}. \end{aligned} \quad (29)$$

Lemma 6.3 (joint core). *For every integer $t \geq 11$ and every real $Y \geq (t-2)(t-7)$, putting $a = Y + 5(t-2)$ gives*

$$\frac{[(2t-1)R_0 - 1]^2}{2aR_0} \geq \frac{27}{20}. \quad (30)$$

Exact polynomial certificate. Clear the positive denominators in (30). For $t = 11, \dots, 16$, the resulting univariate polynomial after $Y = (t-2)(t-7) + Q$ is positive on $Q \geq 0$ by exact Sturm sequences: the sign variations at 0 and $+\infty$ agree and the value at 0 is positive.

For every $t \geq 17$, substitute

$$t = S + 17, \quad Y = Q + (S + 15)(S + 10).$$

The cleared bivariate polynomial has 105 monomials in (Q, S) . Every coefficient is nonnegative and the constant coefficient is positive. It is therefore positive on the whole quadrant $Q, S \geq 0$. The indexed coefficient digest is

9726005502a811c97584bc22b98eeced6577d4b18a5c3c45df3eb178239c69b4.

The two certificates cover all integers $t \geq 11$ and the full real half-line in Y . $\square$

The corridor chain also gives, on (28),

$$\rho(j, t) \geq \frac{27}{20} \left(1 - \frac{2}{4j-3}\right)^2 \left(1 - \frac{2}{j}\right)^2. \quad (31)$$

The two attached factors increase with j . At $j = 504$ their product is

$$\frac{2011}{2013} \frac{251}{252} = \frac{504761}{507276}.$$

Consequently

$$\rho(j, t) \geq \frac{27}{20} \left(\frac{504761}{507276}\right)^2 = \frac{254783667121}{190614029760} = 1.3366469794 \dots > \frac{4}{3}. \quad (32)$$

The row-free core needed to clear $4/3$ after these attached factors is exactly

$$\frac{4/3}{(504761/507276)^2} = \frac{343105253568}{254783667121} = 1.3466532507 \dots,$$

which is strictly below $27/20$.

To verify that (28) lies in the domain of Lemma 6.3, the corridor gives $Y \geq \mathfrak{a}_j - 5(t-2)$. Hence

$$Y > (t-2)^2 - 5(t-2) = (t-2)(t-7).$$

Moreover

$$(t-2)^2 - 5t = (t-9)^2 + 9(t-9) + 4 > 0 \quad (33)$$

for every integer $t \geq 9$, so the original joint hypothesis $\mathfrak{a}_j > 5t$ also holds.

6.6 Coverage proof

Proof of Theorem 6.1. Fix $j \geq t \geq 5$. If $j = t$ or $j = t+1$, use (23). If $t+2 \leq j \leq 503$, use the finite core (24). Thus assume $j \geq 504$.

For $5 \leq t \leq 8$, Table 2 closes the entire column. For $t = 9, 10$, Table 3 and the top floor (27) close the column below the tail, while (25) closes it above the anchor.

Finally let $t \geq 11$. If $\mathfrak{a}_j \leq (t-2)^2$, use the top floor. Otherwise (28) holds, and the joint floor (32) applies. These two inequalities are complementary, with equality assigned to the top side, so no cell is omitted. This exhausts the domain. $\square$

7 Assembly and sharpness

Proof of Theorem 1.1. Proposition 4.1 proves $\Omega_2(r) \geq \omega_{\text{wall}}$, with equality only at $r = 1350$. Proposition 5.1 proves $\Omega_t(r) > 4/3$ for $t = 3, 4$. For $t \geq 5$, Theorem 6.1 and Lemma 3.1 give

$$\Omega_t(r) \geq \min_{t \leq j \leq r} \rho(j, t) > \frac{4}{3}.$$

The exact comparison (7) now gives (5) and shows that no equality is possible away from $(1350, 2)$. Multiplying by 2 proves the raw formulation (6); equality at the stated cell shows that its constant cannot be increased. $\square$

The master join rebuilds ω_{wall} , checks its target fingerprint and its exact comparison with $4/3$, validates the three closed lower-stripe records, every regional certificate and splice, and the symbolic coverage arithmetic. It has 22 named gates; all 22 pass and there are no conditional legs.

7.1 Why monotonicity is not the proof

It is natural to guess that $\Omega_{t+1}(r) \geq \Omega_t(r)$ pointwise. That statement is false. An exact integer comparison at row 45082 gives

$$\Omega_3(45082) < \Omega_2(45082). \quad (34)$$

The witness is the exact integer

$$D = L(G)_{r-3}L(C)_{r-2} - L(G)_{r-2}L(C)_{r-3}, \quad G = (1+z)M_{2r-2}, \quad C = M_{2r-1},$$

whose sign is that of $\Omega_3(r) - \Omega_2(r)$ because $L(C)_q > 0$ on this range. At $r = 45082$ it is negative. Its magnitude has bit length 5,416,154 and 1,630,425 decimal digits, and the SHA-256 digest of its big-endian byte encoding is

$$0\text{aed686f60e5eaf6056db6cb8e1f59aa99610dc26293b7f9f57fb53d2f8ed47e}.$$

Only the top few coefficients of each wall row are consumed, so the companion producer carries a fixed-width top window of the recurrence and reaches $r = 45082$ in well under a minute. It checks that window against the full definition on $5 \leq r \leq 40$ and rejects a recurrence mutation before rebuilding the displayed witness. This does not threaten Theorem 1.1; it only rules out a shortcut. The transfer lemma and the $4/3$ barrier compare each stripe with a common rational floor instead.

The failed low-anchor tail builds have the same status. They identify proof loss in a particular envelope, not counterexamples to the theorem. In particular, the passing build at the largest top-compatible anchor for $t = 9$ shows why a global statement that the envelope route “dies at $t = 9$ ” would be incorrect.

8 A decrement shell and the open profile

The bulk proof exposes a quantitative law that is stronger than the corridor bound needed for Theorem 1.1. In the deposited ladder indexing, put

$$X_{m,k} = 2k(2k+1) \frac{B_k(m)}{B_{k-1}(m)}, \quad d_{m,k} = X_{m,k-1} - X_{m,k}.$$

Proposition 8.1 (certified decrement shell). *For every integer $m \geq 68133$ with $\mathfrak{a}_{m+1} \leq 768$, and every $k \in \{5, \dots, 11\}$,*

$$(k-1)d_{m,k} \leq \frac{6}{5}\sqrt{\mathfrak{a}_{m+1}}. \quad (35)$$

Exact two-leg certificate. The first leg propagates the normalized B – C ladder with 192-bit outward fixed-point intervals. It checks (35) at every row $60000 \leq m \leq 262144$ and all seven indices, with no failure, and deposits the complete interval state at the last row.

For the second leg, write $\beta_k = (\pi/2)B_k$ and use the polynomial flow

$$\beta'_k = \gamma_k, \quad \gamma'_k = \beta_{k-1}.$$

The connection polynomials are the exact closure of this flow, so one interval anchor suffices. Injection errors are majorized in the shifted variable $x = \tau - \tau_{262144}$; the logarithmic tail moments are bounded exactly by

$$\sum_{j>m} \frac{\left(\frac{1}{2} \log(j/m)\right)^i}{j^2} \leq \frac{i!}{2^i m} + \frac{2(i/8)^i}{m^2}.$$

Fifty-seven contiguous windows of width $7/20$ cover $0 \leq x \leq 399/20$. On each window, rational interval subdivision proves the seven cleared inequalities and certifies that every lower envelope which is squared is nonnegative. At the right endpoint the lower bound for $\mathfrak{a}$ exceeds 768; strict increase of $\mathfrak{a}_m$ then closes the stated window. The minimum reported relative margin rises from $0.2719\dots$ to $0.6261\dots$; these decimals report exact rational signs. Retargeting the same certificate from $6/5$ to $4/5$ fails in the first window at six indices, providing a negative control. $\square$

Proposition 8.1 is not needed for the sharp constant: the top–tail join already closes $t = 9$. Combined with the sharpened-corridor slices, however, it gives a second separately gated proof chain for that stripe, parallel to the two chains available for $t = 10$. These routes share the wall definitions and upstream results from [3]; they are not claimed to be fully independent implementations.

The open object is now the profile behind (35), not the displayed inequality itself. Exact data suggest a scaling law of the form

$$d_{m,k} \sim \sqrt{\mathfrak{a}_{m+1}} g\left(\frac{k-1}{\sqrt{\mathfrak{a}_{m+1}}}\right),$$

with the measured normalized decrement near 0.88 on the present window and drifting toward 1. Determining g , proving the limiting constant, and turning the profile into a uniform-in- t deficit transfer remain open. So do the individual higher-stripe minima: the proof uses $4/3$ as a separator, not as their sharp value. Finally, the present results are exact computer-assisted theorems rather than kernel-checked theorems; a proof-assistant replay remains a separate program.

9 Reproducibility and claim boundary

The proof uses symbolic arguments and exact computation in distinct roles. Table 4 states the boundary explicitly.

Table 4: Status of the durable claims.

claim	status	anchor
Theorem 1.1	exact computer-assisted theorem	written transfer and coverage proof plus the 22-gate exact join
terminal and top formulas	all-parameter theorems	symbolic derivations in the text and in [3]
finite core and prefixes	finite exact theorems on stated ranges	integer comparison streams with counts and digests
tail and joint half-lines	exact computer-assisted theorems	rational coefficient signs and exact Sturm sequences
reported decimal capacities	derived reporting values	exact rationals in the certificate records
individual higher-stripe minima	open	exploratory numerical census only
decrement law (35)	exact computer-assisted theorem	192-bit finite ladder, 57 exact envelope windows, and rejected $4/5$ control
limiting decrement profile	open	exploratory scaling data only

No floating-point comparison decides a theorem gate. The finite core uses exact integers; prefix minima use exact cross multiplication; half-line tails use rational interval envelopes and coefficientwise positivity; the joint certificate uses integer polynomials and exact Sturm signs. Every reported decimal is obtained after the exact verdict.

One inherited interval constructor deserves disclosure. Its nominal upper endpoint for π used truncation and therefore produced a singleton lying one fixed-point unit too low. The replay first detects the defect and then repairs the interval by an outward ceiling. Every current consumer immediately scales the interval by $1/2$; exact integer rounding maps the defective and corrected cells to the same interval for $\pi/2$. Thus the repair restores the source contract while changing every downstream coefficient by exactly zero. A hard-coded 100-decimal-digit enclosure for π rejects the old constructor and accepts the repaired one.

The release record contains the exact certificate records, the ordered comparison digest and summary extrema for every prefix stream, the positive-polynomial certificates, the failed-anchor controls, the exact crossing records, and the master join, together with the producer that regenerates each stream in full. The reader-facing source map names the load-bearing artifacts and their digests. This paper makes no kernel-verified claim. The manuscript and its companion replay carry the identifier [doi:10.5281/zenodo.21866366](https://doi.org/10.5281/zenodo.21866366).

10 Conclusion

The bridge constant is controlled by a single exceptional cell, but proving that fact requires a global argument. The direct $t = 2$ certificate finds the cell. The lossless transfer converts the rest of the question into a common $4/3$ square-tail barrier. Terminal identities, a finite core, fixed columns, and the top-joint dichotomy then cover every remaining row without an asymptotic gap.

This separates two constants that play different roles. The factor 2 in the earlier bridge theorem is structural and easy to transport through the Turán triangle. The number 2.655018313913361199726406175676... is extremal: it is the largest raw factor valid over the whole shift wall, and its equality cell is unique.

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Large language models were used as mathematical interlocutors: to test proof routes, carry out derivations, and organize exact certificate searches. Every mathematical claim in the paper is supported by a written proof or an exact certificate, and responsibility for the result is mine.

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